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École polytechnique fédérale de Lausanne
pour l'obtention du grade de Docteur ès Sciences
par

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Lausanne, EPFL, 2026

Acknowledgements

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Lausanne, April 29, 2026

D. K.

Preface

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Lausanne, 12 Mars 2011

T. D.

Abstract

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German abstract

Zusammenfassung

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Résumé

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Introduction

A non-numbered chapter...

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1 Aim and scope

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Chapter 1. Aim and scope

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2 Theoretical background

This chapter introduces the physical and numerical concepts required to interpret the evolution of the inner dark matter density in ultra-faint dwarf galaxy simulations. The objective is not to review all of modern structure formation, but to establish the minimum framework needed for the analyses in subsequent chapters: collisionless halo dynamics, cusp/core phenomenology, numerical heating, gravitational softening, the gravity solver controls used in SWIFT, and the role of shell-to-shell coupling in mass redistribution.

The chapter is intentionally organized from first principles to practical diagnostics. We begin with the collisionless equations that govern halo evolution, then move to the cusp-core problem in faint systems, and finally discuss numerical ingredients that can mimic or amplify physical effects. This progression is important because the central scientific question of the thesis requires separating genuine dynamical evolution from resolution- and solver-induced artefacts.

2.1 Dark matter halos and density profiles

In the collisionless limit, dark matter is described by a phase-space distribution function $f(\mathbf{x}, \mathbf{v}, t)$ that obeys the collisionless Boltzmann equation, coupled to Poisson's equation for the gravitational potential Binney and Tremaine (2008). The key implication is that halo evolution is governed by a smooth mean gravitational field rather than by direct microscopic collisions between particles.

Using standard notation, the collisionless system is governed by

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f - \nabla_{\mathbf{x}} \Phi \cdot \nabla_{\mathbf{v}} f = 0, \quad (2.1)$$

with the potential determined by

$$\nabla^2 \Phi = 4\pi G \rho, \quad \rho(\mathbf{x}, t) = \int f(\mathbf{x}, \mathbf{v}, t) d^3 \mathbf{v}. \quad (2.2)$$

Chapter 2. Theoretical background

These equations make clear that halo structure is a phase-space problem: the density profile cannot be understood independently of the velocity distribution and orbital content.

In cosmological simulations, dynamically relaxed halos are commonly described by broken power-law density profiles, where the logarithmic slope changes from inner to outer radii. In this language, the central shape of the profile is the critical diagnostic: a steep rise toward small radii is referred to as a *cusp*, while a flattened inner profile is referred to as a *core*. Since this inner region is also the most demanding numerically, any physical interpretation must be tied to explicit convergence and resolution arguments Power et al. (2003); Zhang et al. (2019).

A commonly used reference model is the NFW-like form, which captures the transition from an inner and outer power-law regime. The associated local slope,

$$\gamma(r) \equiv -\frac{d \ln \rho}{d \ln r}, \quad (2.3)$$

is a useful quantity throughout this thesis because it directly tracks whether the central profile evolves toward cusp-like or core-like behaviour.

For this thesis, the practical consequence is straightforward: the measured inner density slope cannot be interpreted in isolation. It must be analyzed jointly with the numerical setup that determines force accuracy, relaxation strength, and time integration quality.

2.2 Cusp-core behaviour in ultra-faint dwarf galaxies

Ultra-faint dwarf galaxies are particularly valuable for small-scale dark matter studies because they are strongly dark-matter dominated and have shallow gravitational potentials. Their internal structure is therefore sensitive both to the underlying dark matter model and to dynamical heating processes that can redistribute mass over time Salamin (2026).

In a cold dark matter scenario, cuspy inner profiles are the usual expectation for collisionless halos. In warmer dark matter scenarios, free-streaming suppresses small-scale power and can alter assembly histories, potentially affecting central structure indirectly through different accretion and merger pathways Salamin (2026). Independently of the particle model, repeated perturbations and non-adiabatic potential fluctuations can transfer energy to central orbits and drive profile flattening.

The conceptual point is that core formation is not a binary signature of one mechanism. Similar-looking inner profiles can emerge from different routes: altered initial fluctuation spectra, stochastic heating by external perturbers, or numerical relaxation. As a result, robust interpretation requires combining profile evolution with complementary diagnostics such as enclosed-mass evolution, shell-resolved kinematics, and sensitivity to solver controls.

This creates a central interpretive challenge for the present work: an apparent cusp-to-core transition can arise from physical mechanisms, from numerical artifacts, or from both acting

simultaneously. The role of the theoretical background is therefore to define diagnostics that separate these possibilities as rigorously as possible.

2.3 Numerical heating and relaxation effects in N -body systems

An N -body simulation represents a continuous collisionless fluid with a finite number of massive macro-particles. This discretization introduces spurious collisionality: close encounters can produce artificial two-body relaxation, velocity diffusion, and long-term heating, especially in poorly resolved central regions Binney and Tremaine (2008); Power et al. (2003).

To first order, the local relaxation timescale scales approximately as

$$t_{\text{relax}} \sim \frac{N}{8 \ln N} t_{\text{cross}}, \quad (2.4)$$

where N is the number of effective particles supporting the region of interest and t_{cross} is a characteristic crossing time Binney and Tremaine (2008). This scaling emphasizes why central halo regions are vulnerable: they contain fewer particles and shorter dynamical times, making discreteness effects comparatively stronger.

The classical relaxation timescale argument provides a first-order test: if the numerical relaxation time at a given radius is not much longer than the simulated evolution time, then the resulting structure may be numerically biased. In practice, this means that central density trends must always be interpreted together with particle number, timestep control, force regularization, and force-accuracy parameters.

Power et al. translated these ideas into practical convergence guidance for halo simulations by linking reliability in the inner profile to a combination of mass resolution, timestep sampling, and force softening Power et al. (2003). Their framework remains a benchmark because it moves beyond single-parameter tuning and treats convergence as a coupled numerical problem.

For this thesis, numerical heating is not treated as a generic warning but as an explicit hypothesis to test. If central density decline is dominated by relaxation artifacts, it should correlate with changes in numerical controls. If it persists across broad parameter changes, a more global dynamical mechanism becomes more plausible.

2.4 Gravitational softening: fixed and adaptive formulations

Gravitational softening modifies the short-range Newtonian force to prevent unphysical singular accelerations from particle-particle encounters. With a fixed softening length ϵ , one obtains a trade-off: large ϵ suppresses noise and relaxation but erases genuine small-scale structure; small ϵ improves nominal resolution but can increase discreteness-driven heating Power et al. (2003); Zhang et al. (2019).

Chapter 2. Theoretical background

In many implementations, this can be written schematically with a softened pair force,

$$\mathbf{F}_{12} = -\frac{Gm_1m_2}{(r_{12}^2 + \epsilon^2)^{3/2}} \mathbf{r}_{12}, \quad (2.5)$$

which regularizes the r^{-2} singularity at very short separation Zhang et al. (2019). The interpretation of ϵ is therefore two-fold: it is both a numerical stabilizer and a scale that limits physically meaningful force gradients.

Power et al. established influential convergence criteria linking softening, timestep, and particle number in the inner halo Power et al. (2003). Later work showed that these prescriptions may be conservative for some setups, while also emphasizing that excessively small softening can bias the innermost density high Zhang et al. (2019). Taken together, these studies show that softening is not a single "best" number independent of context.

Adaptive softening addresses this by varying effective force resolution with local conditions. However, naive implementations violate conservation laws. The Lagrangian formalism of Price and Monaghan demonstrated how to construct adaptive softening with the additional correction terms required for consistent energy and momentum conservation Price and Monaghan (2007). Modern conservative extensions generalize this framework to multi-species collisionless simulations Hopkins et al. (2023).

From a methodological perspective, adaptive schemes are especially relevant for halos with strong radial density contrasts. A fixed ϵ that is adequate in the outskirts may be too coarse in the center, while an ϵ tuned to the center may be too noisy at large radii. Conservative adaptive formulations provide a way to mitigate this tension while preserving the invariants required for long integrations.

In the context of this thesis, softening is a key control variable, but not an isolated one. Its interpretation depends on the interaction between local force regularization and global gravitational coupling across the halo.

2.5 Gravity solver controls in SWIFT: Tree/FMM context

Beyond softening, force computation in modern cosmological codes depends on hierarchical gravity solvers (tree and fast multipole expansions) and their acceptance criteria. These methods accelerate long-range gravity by replacing exact pairwise summation with controlled approximations, whose errors depend on opening criteria and expansion tolerances.

This introduces a second numerical layer on top of softening. Even with identical initial conditions and identical softening, changes in opening criteria or multipole tolerances can alter force errors in a way that accumulates over many dynamical times. For slowly evolving central densities, small systematic differences can integrate into measurable secular trends.

2.6 Why shell structure and mass redistribution matter

In practical terms, parameters such as the multipole acceptance criterion (MAC), geometric opening thresholds (e.g. θ_{cr}), expansion tolerances (e.g. FMM error controls), and timestep factors set the balance between computational cost and force accuracy. Because central density evolution is sensitive to small cumulative force biases, these controls must be included in any robust sensitivity analysis of cusp/core evolution.

The same logic applies to timestep controls. If timesteps are too large compared to local orbital times, integration error can mimic diffusion or heating. If they are too small, the simulation becomes expensive but not necessarily more accurate unless force errors are also reduced. A reliable interpretation therefore requires coordinated tests across both integration and force-approximation parameters.

Therefore, when this thesis tests solver settings, the goal is not merely code tuning. It is a mechanism test: if the observed inner-density trend is primarily a solver artifact, changing these controls should alter either its onset or its long-term rate.

2.6 Why shell structure and mass redistribution matter

A central point for this thesis is that inner density evolution is not purely local. Even when interest focuses on the innermost region, the central potential is influenced by mass and orbital structure over a broad radial range. Particle populations at larger radii can contribute to central acceleration fluctuations, angular momentum exchange, and gradual redistribution of enclosed mass.

Recent theoretical work on stochastic tidal fields generated by clumpy substructure provides a useful interpretation for this non-local coupling. In these treatments, gravitational fluctuations induce diffusion in orbital integrals and can transfer energy and angular momentum through repeated perturbations rather than through single catastrophic events Peñarrubia (2019, 2017). This view is directly relevant to shell-based diagnostics: outer shells can influence central dynamics through cumulative, time-correlated forcing.

An important implication is that central density decline may be driven by the full halo environment, not only by the innermost force law. If true, changing local softening alone may leave a substantial part of the trend unchanged. This is precisely why shell-resolved measurements and cumulative mass diagnostics are central to the thesis design.

This motivates a shell-based analysis in which the halo is decomposed into radial regions and their dynamical influence on the center is tracked over time. If outer shells contribute substantially to central kinematic evolution, then changing only inner local parameters (for example, local softening) may have limited impact on the global trend.

This non-local perspective is consistent with previous findings in the project context, where persistent central density decline can remain visible across multiple numerical variations Salamin (2026). The theoretical implication is that diagnosis must combine local resolution

tests with global mass-transport diagnostics.

2.7 Chapter summary

The theoretical framework developed here leads to three working principles for the rest of the thesis. First, cusp/core evolution in ultra-faint dwarfs is physically informative only when accompanied by explicit numerical convergence and solver-sensitivity checks. Second, gravitational softening is necessary but not sufficient to explain long-term inner-density evolution. Third, shell-to-shell coupling and cumulative mass redistribution provide a natural mechanism class that must be tested alongside local numerical effects.

Operationally, this means that the results chapter should be read as a sequence of discriminating tests. Each test changes one class of assumptions and asks whether the central trend persists, weakens, or disappears. A persistent trend across softening and solver variations strengthens the case for a global redistribution process; strong sensitivity to one parameter would instead indicate a dominant numerical origin.

The next chapter translates these principles into a concrete simulation and analysis framework, including parameter studies, centering strategy, density diagnostics, and shell-based measurements.

3 Simulation setup, framework and implementation

This chapter describes the numerical setup and analysis pipeline used to investigate the persistent central density decline identified in the previous chapters. The focus is on reproducibility: each subsection documents the simulation code configuration, initial-condition choices, diagnostic definitions, and implementation details used to test competing numerical and physical explanations.

The workflow combines (i) controlled collisionless N -body runs in SWIFT, (ii) systematic parameter variations of the self-gravity solver, (iii) profile and cumulative-mass diagnostics extracted from snapshots, and (iv) shell-based analyses designed to quantify non-local contributions to the dynamics of the innermost region.

3.1 SWIFT simulation setup

All simulations in this thesis are dark-matter-only and non-periodic, evolved with SWIFT in self-gravity mode (and external gravity when required by specific tests). In the working setup, runs are launched through

```
swift --self-gravity [--external-gravity] params.yml
```

with outputs consisting of snapshots, global statistics, and timestep logs.

The internal unit system is chosen so that

$$1 \text{ length unit} = 1 \text{ kpc}, \quad 1 \text{ velocity unit} = 1 \text{ km s}^{-1}, \quad 1 \text{ mass unit} \approx 10^{10} M_{\odot}, \quad (3.1)$$

which allows straightforward interpretation of profile radii (kpc), velocity dispersions (km/s), and enclosed masses in solar units after post-processing conversion.

Chapter 3. Simulation setup, framework and implementation

3.1.1 Baseline gravity and time integration choices

The parameter-study baseline adopts the self-gravity configuration

$$\eta = 0.002, \quad \text{MAC} = \text{geometric}, \quad \theta_{\text{cr}} = 0.7, \quad \epsilon_{\text{fmm}} = 10^{-3}, \quad (3.2)$$

with minimum and maximum timesteps typically constrained in the interval

$$\Delta t_{\text{min}} = 10^{-6}, \quad \Delta t_{\text{max}} = 10^{-2} \quad (3.3)$$

for the long-run campaign, and tighter values used in dedicated tests.

To probe solver sensitivity, each of the following was varied while keeping the others fixed: timestep factor η , MAC choice (geometric/adaptive), geometric opening threshold θ_{cr} , and ϵ_{fmm} . These controlled perturbations define the backbone of the numerical-robustness analysis presented in the results chapter.

3.1.2 Output cadence and bookkeeping

The production analysis relies on regular snapshot dumps that allow profile reconstruction at multiple times (e.g. 0, 20, 40, 60, 80, and 100 Gyr in the long runs). In parallel, SWIFT writes:

- global conservation and bulk quantities in `statistics.txt`,
- timestep activity in `timesteps.txt`.

These files are used as consistency checks (mass budget, energy trends, and global angular momentum evolution) when interpreting local density changes.

3.2 Halo initial conditions and parameter choices

The halo models are generated as collisionless particle realizations with controlled inner slopes to enable cusp/core comparisons under otherwise matched numerical conditions.

For dedicated truncated-NFW tests, initial conditions are produced with

```
ic_gen_2_slopes --alpha 1 --beta 3 --Rmin 1e-3 --Rmax 50 --r_cutoff 50 \  
--rho0 0.0027658297046461644 --rs 0.828177997155062 \  
--ptype 1 --mass 61664.060524 -t swift -o halo.hdf5
```

which corresponds to a two-slope halo with tunable inner structure and an outer truncation radius.

Across the broader thesis campaign, the most relevant controlled parameters are:

- halo type: cusp and core initial conditions,
- box-size variants for finite-volume sensitivity checks,
- gravitational softening variants (with a commonly used reference value $\epsilon = 0.01$ kpc),
- self-gravity solver parameters (η , MAC, θ_{cr} , ϵ_{fmm}).

The practical philosophy is to keep one baseline run and modify one parameter family at a time, so each subsection in the results can be interpreted as a direct hypothesis test rather than a multi-parameter confounded comparison.

3.3 Density, shell, and mass diagnostics

This section defines the observables used throughout the thesis.

3.3.1 Spherically averaged density profiles

Density profiles are extracted from snapshots with a radial grid up to a user-defined r_{max} and N_r bins. In practice, the analysis scripts use both moderate and high radial resolutions, including $N_r = 500$ in the inner-region-focused runs.

For shell i with mass ΔM_i and boundaries $(r_{i,\text{in}}, r_{i,\text{out}})$, the estimator is

$$\rho_i = \frac{\Delta M_i}{\frac{4\pi}{3} (r_{i,\text{out}}^3 - r_{i,\text{in}}^3)}. \quad (3.4)$$

Profiles are then compared to the initial state using ratio curves $\rho(r, t)/\rho(r, t_0)$, which makes secular flattening or depletion immediately visible.

3.3.2 Softening-scale marker in profile plots

To separate physically interpretable radii from numerically delicate scales, profile plots include a vertical reference at

$$r_{\text{ref}} = 2.8\epsilon, \quad (3.5)$$

with ϵ the run softening length. This marker is not used as a hard cutoff, but as a visual guide when discussing central profile evolution.

3.3.3 Cumulative mass profiles

Cumulative mass diagnostics are computed via

$$M(< r) = \sum_{r_j < r} m_j, \quad (3.6)$$

Chapter 3. Simulation setup, framework and implementation

including optional decomposition by particle type when needed. The implementation supports multiple centering modes (center of mass, median, manual center, and shrinking-sphere center), with shrinking-sphere used in the dedicated cumulative-mass tests because of its robustness to asymmetries and mild off-centering.

3.3.4 Velocity and anisotropy diagnostics

Using the same radial binning, the analysis computes velocity moments and anisotropy indicators, including

$$\beta(r) = 1 - \frac{\sigma_t^2}{2\sigma_r^2}, \quad (3.7)$$

as complementary diagnostics for identifying orbital heating and redistribution mechanisms.

3.4 Shell-based analysis framework

To test whether the central evolution is driven purely by local inner-halo physics or by non-local coupling, the halo is decomposed into radial shells and each shell's contribution to central perturbations is quantified.

The shell study scripts use logarithmically spaced shells between inner and outer radii and evaluate a proxy for the velocity-perturbation contribution,

$$\Delta V_{\text{shell}}^2 \propto \frac{16\pi G^2 m_p}{v_{\text{rel}}^2} \int_{r_{\text{in}}}^{r_{\text{out}}} \rho(r) dr, \quad (3.8)$$

with m_p the particle mass and v_{rel} a characteristic relative velocity scale in the model.

For an NFW-like profile, this integral is evaluated analytically in the implementation, enabling direct shell-by-shell comparisons without introducing numerical integration noise at this stage.

The practical output is a radial contribution spectrum that indicates how much outer shells can influence central kinematics. This framework is central to the thesis logic because it directly tests the hypothesis that central decline may be maintained by global halo coupling even when local numerical parameters are modified.

3.5 Numerical implementation of the hybrid model

Motivated by the shell results, a dedicated hybrid gravity model was implemented and tested on a custom SWIFT branch (truncated-NFW). The model enforces a hard radial split at a cutoff radius R_{cut} :

3.6 Centering, snapshot concatenation, and noise control

- inner region ($r \leq R_{\text{cut}}$): standard self-gravity, no external NFW acceleration,
- outer region ($r > R_{\text{cut}}$): external NFW acceleration applied, with optional masking of dark-matter self-gravity sourcing outside the cutoff.

In test runs, the backend is selected at compile time using

```
--with-ext-potential=truncated-nfw
```

and runtime control is provided through the `NFWPotential` block, including

$$R_{\text{cut}} = 10 \text{ kpc}, \quad \text{dm_mask_outside_cutoff} = 1 \quad (3.9)$$

in the reference validation setup.

The design goal is not to claim a final physical model, but to construct a targeted numerical experiment: if outer-halo coupling is a major driver of central evolution, changing how the outer region sources gravity should produce a measurable response in inner-density trends.

3.6 Centering, snapshot concatenation, and noise control

Because the central-density signal is subtle and cumulative, robust post-processing is essential. Three practical steps are applied:

- explicit centering checks (including histogram/peak and shrinking-sphere based workflows),
- consistent recentering in profile extraction scripts,
- snapshot concatenation to reduce shot noise in long-run diagnostics.

For the 100 Gyr campaign, concatenated datasets were built in windows of 10 snapshots (100 Myr effective spacing) and 50 snapshots (500 Myr windows). This improves visual stability of profile curves and helps distinguish persistent trends from per-snapshot noise fluctuations.

Importantly, chunking is treated as a representation-control tool rather than a physical correction: reductions in noise do not by themselves resolve secular central-density decline, so interpretations remain anchored to the full diagnostic set.

3.7 Chapter summary

The methodological framework of this thesis combines controlled simulation design and targeted diagnostics:

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- controlled SWIFT parameter studies of softening and gravity-solver settings,
- matched cusp/core and box-size comparisons,
- profile, cumulative-mass, and kinematic diagnostics under consistent centering,
- shell-based non-local coupling analysis,
- and a dedicated truncated-NFW hybrid implementation test.

This setup is designed so that each result in the next chapter can be interpreted as a specific mechanism test, progressively ruling out or supporting candidate explanations for the observed central-density evolution.

4 Results and discussion

This chapter presents the numerical results obtained from the reference simulation campaign and the subsequent sensitivity studies. The central question is whether the persistent decline of the inner density is an intrinsic dynamical feature of the halo, a consequence of numerical settings, or an artefact of the analysis procedure. The discussion is therefore structured around a sequence of increasingly restrictive checks: first the reference evolution itself, then the post-processing pipeline, then controlled variations of the simulation volume, the initial halo structure, the softening length, and the gravity solver settings.

The completed sections below are written as a scientific discussion rather than as a notebook-style log. The later sections remain intentionally unfinished and are marked with explicit placeholders.

4.1 Baseline behaviour

The baseline run shows a clear cusp-to-core transition, but the core does not stabilize. Instead, the inner density continues to decline throughout the simulated time span. This is the central empirical result of the thesis. In the broader context of halo evolution, the literature shows that collisionless systems can display a wide range of inner responses to external perturbations and time-dependent forcing, including stochastic heating by substructure and repeated diffusion in energy–angular-momentum space Peñarrubia (2017, 2019). The reference run is therefore notable not because it forms a core, but because the core remains dynamically active rather than approaching a steady inner state.

The result is also instructive from the point of view of numerical convergence. Power et al. showed that the inner halo is especially sensitive to the interplay between resolution, force softening, and timestep control Power et al. (2003), while Zhang et al. later demonstrated that too aggressive a reduction of the softening length can artificially raise the innermost density Zhang et al. (2019). The reference simulation does not display a simple monotonic dependence on any one of these parameters, which already suggests that the behaviour is not

Chapter 4. Results and discussion

a trivial one-parameter resolution effect.

The baseline density-evolution plot and the snapshot sequence show the same qualitative pattern: the inner profile becomes shallower, but the inner density does not settle. In that sense, the evolution resembles a continuing redistribution of mass rather than a completed structural transformation. This is the key point that motivates the rest of the chapter.

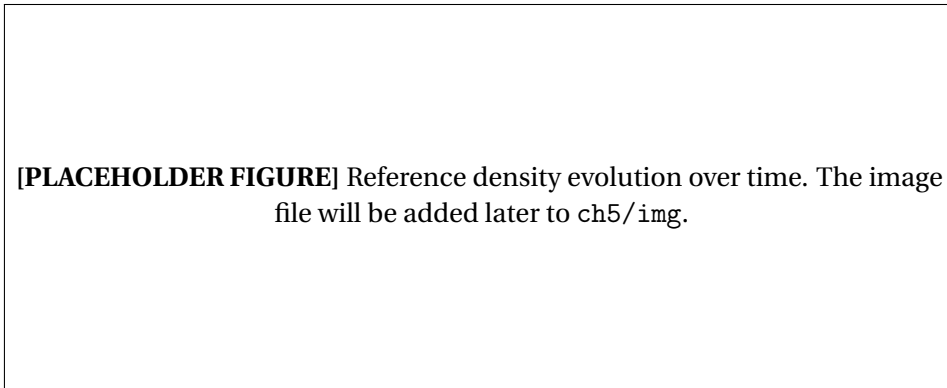


Figure 4.1: [PLACEHOLDER FIGURE] Reference density evolution over time. The image file will be added later to ch5/img.

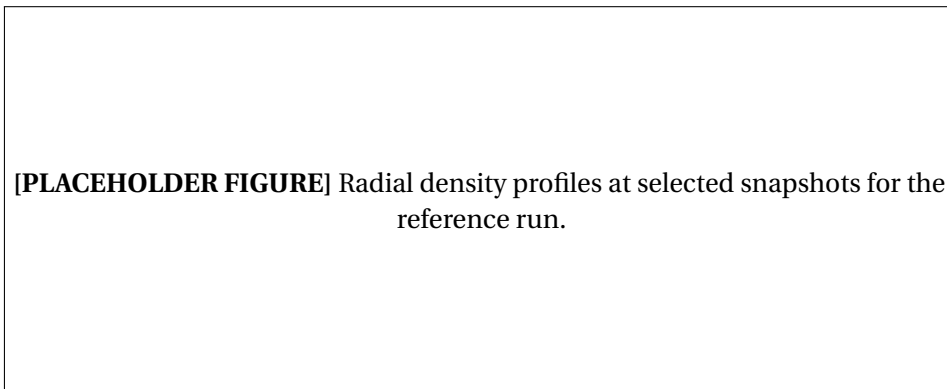


Figure 4.2: [PLACEHOLDER FIGURE] Radial density profiles at selected snapshots for the reference run.

The comparison with the literature is therefore important: the simulation reproduces a cusp-to-core transition, but it does not reproduce a long-lived, stationary core. Whatever mechanism is responsible must therefore operate over the full duration of the run.

The most conservative conclusion at this stage is that the halo does not relax into a static inner structure on the simulated timescale. The observed behaviour is a secular evolution of the center, not merely an initial transient.

4.2 Analysis pipeline

Because the effect is subtle and cumulative, the first concern is whether it might be produced by the analysis pipeline rather than by the dynamics itself. Two sources of bias are particularly relevant here. The first is centering: if the halo center drifts relative to the coordinate system, the innermost density can be artificially suppressed. The second is snapshot noise: individual outputs from an N -body simulation can fluctuate enough that a short-lived dip looks like a trend if the signal is sampled too sparsely.

The centering issue was addressed by modifying the post-processing scripts to recenter each snapshot using a shrinking-sphere method. In addition, the original snapshot sequence was concatenated into larger time blocks, first with 10-snapshot chunks and then with 50-snapshot chunks. This is a useful check because concatenation reduces the variance associated with single snapshots while preserving secular evolution.

The results show that both operations improve the appearance of the curves, but neither changes the underlying trend. The recentered profiles are consistent with the original analysis, and the chunked curves remain systematically declining even though they are much smoother. That is an important distinction: the post-processing pipeline affects the noise level, but not the direction of the evolution.

This is also where the density-evolution plots become easier to interpret physically. A noisy sequence can obscure whether the core is stabilizing, but if stronger temporal smoothing still yields the same declining envelope, then the effect is not an artefact of single-snapshot fluctuations. The signal survives the most obvious analysis-level corrections.

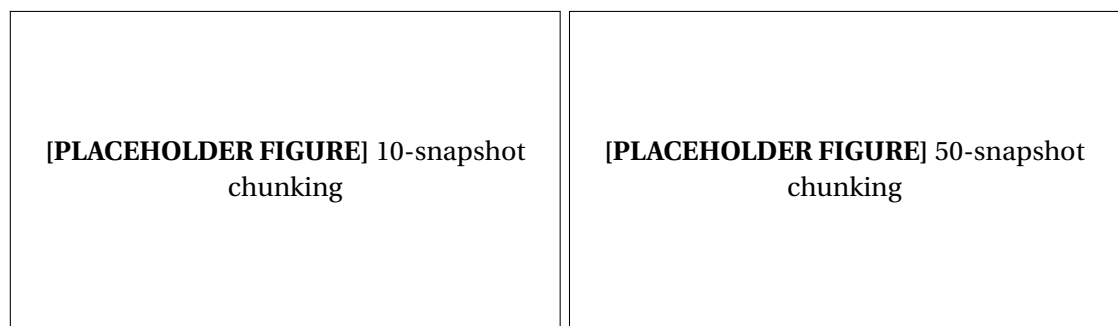


Figure 4.3: **[PLACEHOLDER FIGURE]** Comparison of 10-snapshot and 50-snapshot concatenated profiles. The exact figure filenames will be added later in `ch5/img`.

The conclusion is straightforward: the central decline is robust against the two most plausible post-processing artefacts, namely centering drift and snapshot noise. The pipeline matters for presentation, but not for the existence of the trend itself.

[PLACEHOLDER FIGURE] Comparison of centering methods, including shrinking-sphere and histogram-based centering.

Figure 4.4: [PLACEHOLDER FIGURE] Comparison of centering methods, including shrinking-sphere and histogram-based centering.

4.3 Box size

The dependence on box size is more interesting. Increasing the box size appears to slow down the decline, although the effect is modest rather than dramatic. In the current notes, the reference comparison is approximately a drop to ~ 0.6 of the initial central density for the 100-box run versus ~ 0.75 for the 500-box run after 10 Gyr. That difference is large enough to matter, but not large enough to rewrite the overall interpretation.

One way to think about this result is that the outer domain contributes to the gravitational environment felt by the center, but only weakly changes the qualitative behaviour. This is consistent with the broader discussion in the literature on stochastic perturbations by substructure: long-range fluctuations can alter the rate of evolution without necessarily determining the existence of the effect itself Peñarrubia (2017, 2019).

The box-size study therefore suggests that the central evolution is not a purely local numerical issue. If the entire behaviour were controlled by the immediate neighborhood of the center, one would expect box size to matter only weakly. Instead, the result indicates a modest environmental dependence. In practical terms, this means that the volume size may regulate the timescale of the decline, but not its existence.

The figure set for the box-size study should therefore be read as a sensitivity check rather than a decisive mechanism test. It tells us that the observed decline is not insensitive to the global numerical domain, but it also shows that the phenomenon is not eliminated by enlarging the box.

4.4 Initial halo structure

The comparison between cuspy and cored initial conditions is particularly informative because it distinguishes between a cusp-specific instability and a more general process. The core run

[PLACEHOLDER FIGURE] Density evolution for different box sizes.

Figure 4.5: **[PLACEHOLDER FIGURE]** Density evolution for different box sizes.

[PLACEHOLDER FIGURE] Ratio plot or summary diagnostic for box-size sensitivity.

Figure 4.6: **[PLACEHOLDER FIGURE]** Ratio plot or summary diagnostic for box-size sensitivity.

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still exhibits a decline of the central density, but the amplitude is smaller than in the cuspy case. This is a strong result because it means the mechanism is not limited to the special case of an initially steep central profile.

In physical terms, this suggests that the system is sensitive to the presence of a central potential well, but not exclusively to its initial logarithmic slope. A cored halo has less extreme inner structure and therefore reacts less strongly, yet the evolution remains qualitatively the same. That is exactly what one would expect if the central response is driven by a process that acts through the halo as a whole rather than through a cusp alone.

The literature on substructure-driven heating is again relevant here. Stochastic forcing does not require a cusp; it requires a fluctuating potential and a population of perturbers that can exchange energy with the central region Peñarrubia (2017). The persistence of the effect in the core run is therefore compatible with a non-local heating picture and inconsistent with the idea that the problem is merely a cusp pathology.

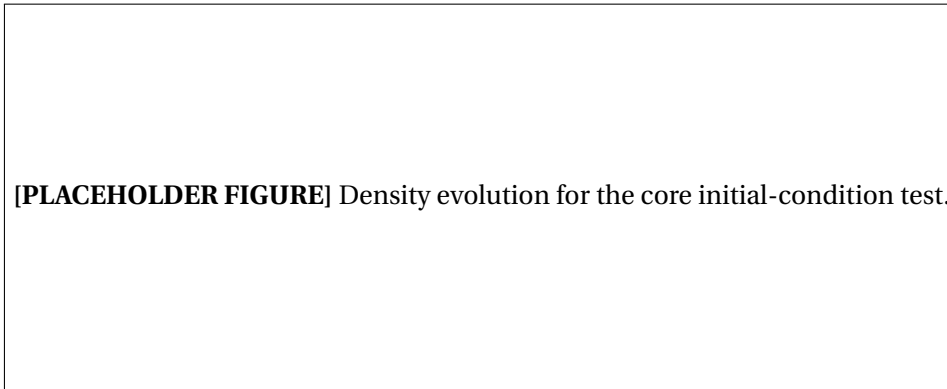


Figure 4.7: [PLACEHOLDER FIGURE] Density evolution for the core initial-condition test.

The practical implication is that changing the inner slope reduces the amplitude but not the mechanism. The central decline is therefore not restricted to cuspy initial conditions.

4.5 Gravitational softening

Softening is the most obvious numerical parameter to suspect in any study of the inner halo, because it directly regularizes short-range interactions. The literature makes clear, however, that the role of softening is subtle: too much softening suppresses genuine small-scale structure, while too little can artificially increase the density in the innermost region Power et al. (2003); Zhang et al. (2019). The present results fit that picture in the sense that the softening length changes the detailed rate of evolution, but not the existence of the decline itself.

This is an important negative result. If softening were the primary cause, the central density evolution should have shown a qualitatively different response when the softening length was

varied. Instead, the trend persists across the tested range. That makes the softening study particularly useful because it rules out a very common explanation without requiring any additional assumptions.

The notes from the shell analysis already point in the same direction: outer shells contribute significantly to the central dynamics, and changing the local softening length does not alter the fact that those shells are still present and still exert gravity. In other words, softening can change the numerical regularization of pairwise forces, but it does not remove the mass distribution that sources the global potential.

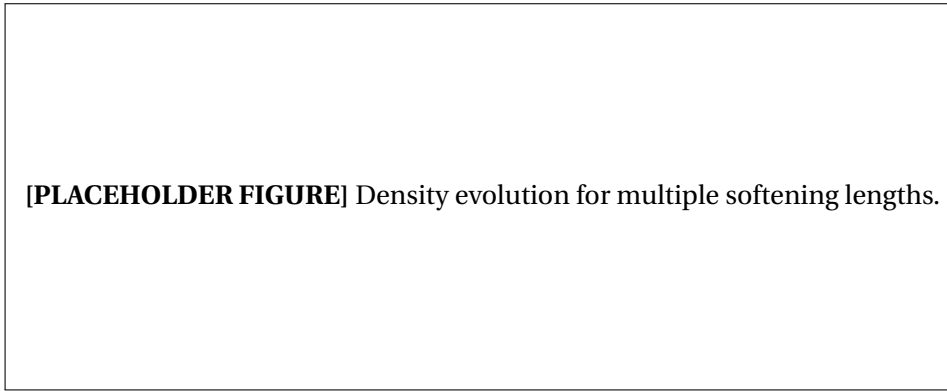


Figure 4.8: [PLACEHOLDER FIGURE] Density evolution for multiple softening lengths.

The conclusion is that softening is not the dominant driver. It is part of the numerical control budget, but not the mechanism behind the secular central depletion.

4.6 Self-gravity settings

The solver-parameter study addresses a more subtle question: whether the observed decline could be the cumulative effect of force-approximation or integration errors. This is a natural concern in hierarchical gravity solvers, where the acceptance criterion and multipole tolerances determine how accurately long-range interactions are represented. In principle, a slightly looser or tighter force approximation can shift the secular evolution of a halo even when the initial conditions are identical.

The present study varies four key control parameters of the self-gravity solver:

- **Timestep factor η :** Controls the integration timestep via $\Delta t = \eta \times t_{\text{dyn}}$. A smaller η means finer temporal resolution and more frequent force updates; a larger η allows fewer timesteps and can accumulate more integration error over each step.
- **Multipole Acceptance Criterion (MAC):** Determines when the force from a distant cell can be approximated by its multipole moments rather than evaluated exactly from individual particles. The *geometric* MAC uses the cell size and distance; the *adaptive*

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MAC adjusts the criterion based on the local particle distribution. Geometric MAC is faster but less accurate for clumpy structures.

- **Geometric opening angle θ_{cr} :** Specifies the threshold for geometric MAC: cells with $\theta_{\text{cr}} = r_{\text{cell}}/d < \theta$ activate multipole approximation. Smaller θ_{cr} means a stricter criterion, forcing more exact evaluations and improving accuracy at higher computational cost.
- **FMM tolerance ϵ_{fmm} :** Controls the precision of the Fast Multipole Method used to compute far-field forces. Smaller tolerances mean higher precision, but smaller ϵ_{fmm} requires more multipole terms and increases computation time.

The parameter range explored here is broad enough to reveal systematic sensitivity if it exists. The timestep factor was varied by roughly an order of magnitude, the MAC was changed between geometric and adaptive formulations, θ_{cr} was tightened in the geometric case, and ϵ_{fmm} was varied over two to three orders of magnitude. In a purely numerical explanation, one would expect at least one of these settings to have a decisive effect on the inner profile.

4.6.1 Timestep study

The timestep factor η determines the frequency of gravity and kinematics updates. Increasing η allows larger timesteps and reduces the number of force evaluations per unit time; decreasing η enforces finer temporal sampling and can improve phase-space accuracy. The baseline value of $\eta = 0.002$ is typical for high-resolution dark-matter studies, but it is instructive to test both more conservative ($\eta = 0.0002$, roughly $10\times$ finer) and more permissive ($\eta = 0.02$, roughly $10\times$ coarser) variants.

If timestep error were the main driver of central decline, the coarser run should exhibit notably faster or more severe depletion, while the finer run might show stabilization. That is not observed.

4.6.2 Multipole Acceptance Criterion study

The MAC choice reflects a fundamental trade-off in hierarchical tree methods: speed versus local accuracy. The geometric MAC is computationally cheaper but treats all distant cells identically; the adaptive MAC is more expensive but adapts the multipole threshold based on local structure. For a collisionless halo, the choice of MAC affects the accuracy of long-range forces and can subtly alter how external shells couple to the center.

If MAC were the primary sensitivity, changing between geometric and adaptive should produce a qualitatively different central-decline rate. The runs tested here show no such distinction.

4.6.3 Geometric opening angle study

For the geometric MAC, a smaller opening angle θ_{cr} means stricter multipole approximation: more cell pairs are forced into exact evaluation rather than multipole summation. The baseline value $\theta_{\text{cr}} = 0.7$ is a standard choice; the test value $\theta_{\text{cr}} = 0.5$ represents a more stringent criterion and therefore higher force accuracy.

Lowering θ_{cr} should increase computational cost but also increase force precision. If force-approximation error were limiting the simulation, tightening θ_{cr} should reduce or delay the central decline. The observed effect is marginal.

4.6.4 FMM tolerance study

The Fast Multipole Method tolerance ϵ_{fmm} is a global precision parameter for the multipole expansion. The baseline value is $\epsilon_{\text{fmm}} = 0.001$ (relatively tight), and the test suite includes both looser values (0.01 and 0.1, representing $10\times$ and $100\times$ relaxation) and a tighter value (0.0001, representing $10\times$ tightening). This range is designed to unambiguously reveal whether force approximation is a controlling factor.

Some runs with modified ϵ_{fmm} exhibit slightly improved early-time behaviour, particularly the tighter tolerance, but even those runs exhibit long-term central decline. The change is not enough to alter the qualitative long-term trend.

4.6.5 Summary table of self-gravity parameter variations

Table 4.1: Summary of self-gravity solver parameter values in the study. Each row lists the parameter, its baseline value (used in the reference run), and the test values explored to assess sensitivity.

Parameter	Baseline	Test values		
η (timestep)	0.002	0.0002	0.02	—
MAC	geometric	adaptive	—	—
θ_{cr} (geometric only)	0.7	0.5	—	—
ϵ_{fmm} (FMM tolerance)	0.001	0.0001	0.01	0.1

4.6.6 Overall interpretation of solver sensitivity

That is not what we observe across the full parameter set. The central density evolution remains broadly similar across all tested settings. This is consistent with the general lesson from convergence studies: changing a gravity control parameter may reduce error bars or shift the onset of bias, but it does not necessarily eliminate the physical or numerical process that dominates the evolution Power et al. (2003); Zhang et al. (2019).

The most useful interpretation is therefore not that the solver settings are irrelevant, but that

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they are not the principal explanation. Their influence is secondary and limited to rate-level effects rather than qualitative behaviour. A solver parameter would need to have produced a qualitatively different central decline — either eliminating it or dramatically accelerating it — to be considered a dominant factor. None of the parameters tested here achieve that threshold.

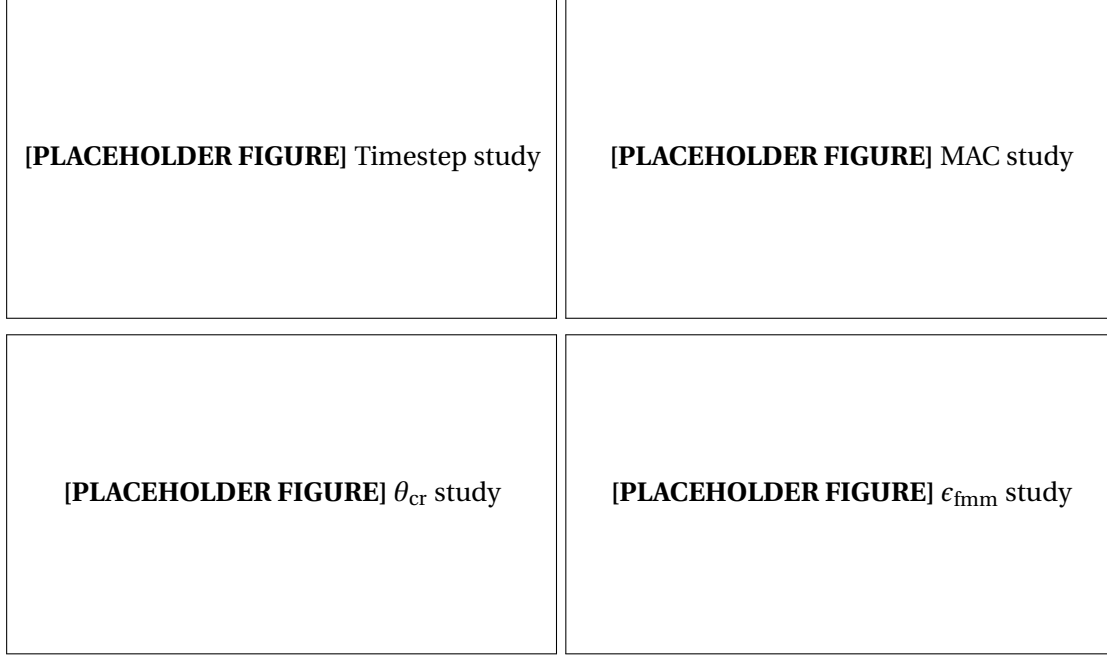


Figure 4.9: [PLACEHOLDER FIGURES] Parameter-study panels for the self-gravity settings. The final image files will be placed in `ch5/img`.

The conclusion is that the trend is robust against the standard solver controls. This removes another plausible numerical explanation and strengthens the case for a mechanism tied to the structure of the halo rather than to a single gravity-accuracy parameter.

4.7 Cumulative mass and angular momentum

This section is not yet completed.

[PLACEHOLDER RESULT] Here we will quantify whether mass is moving outward from the center, whether $M(< r)$ decreases at small radii, and whether the central angular momentum increases over time.

[PLACEHOLDER FIGURE] Cumulative mass enclosed within radius as a function of time.

[PLACEHOLDER FIGURE] Mass-profile evolution at several snapshots.

[PLACEHOLDER FIGURE] Central angular momentum versus time.

4.8 Shell study

This section is not yet completed.

[PLACEHOLDER RESULT] Here we will present the shell-contribution analysis and show why the outer shells matter for the central particle's dynamics.

[PLACEHOLDER FIGURE] Shell contribution to velocity dispersion over time.

[PLACEHOLDER FIGURE] Cumulative contribution by radius.

4.9 Truncated NFW

This section is not yet completed.

[PLACEHOLDER RESULT] Here we will compare the standard runs against the truncated-NFW hybrid model.

[PLACEHOLDER FIGURE] Baseline versus truncated-NFW central-density evolution.

4.10 Synthesis

This section is not yet completed.

[PLACEHOLDER RESULT] Here we will summarize which explanations have been ruled out and which remain plausible.

[PLACEHOLDER FIGURE] Final synthesis plot or compact summary table of the tests.

5 Conclusions

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Chapter 5. Conclusions

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A An appendix

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